

## Inputs

### Building stock

footprints, archetypes,  
dwelling counts

### Costs and performance

capital, O&M, efficiencies,  
lifetimes, discount rate

### Power and geology

2050 capacity, renewables,  
salt caverns

## Heat demand

A 1,369-region NUTS3 residential useful-heat demand surface reconstructed bottom-up across 29 markets and validated against national statistics

## Scenarios

200 Monte Carlo draws over 2025 to 2050, under Current Policies, Stated Policies, Net Zero and H2 Push

## Three tests, applied in order

### 1. Levelised cost

Which carrier gives a MWh of useful heat for least, each charged its last mile and its seasonal store?

### 2. Merit order

Which is cheapest to run at the winter peak, across building heat, district heat and the power peaker?

### 3. Capital recovery

Does scarcity rent cover the annualised cost of building the plant, and if not, what payment would?

## Outputs

Cost gaps and technology shares by country, emissions pathways, and the capacity payment a hydrogen peaker would need to be built